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Linking Neuronal Genetic Identity to Connectivity Patterns in the Entorhinal Cortex

by

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Supervised by
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Executive Summary

The mammalian brain is a highly complex and multifunctional organ, and the entorhinal cortex plays an important role in memory processing. This study investigates the relationship between gene expression and brain connectivity, focusing on how cell-body position within the entorhinal cortex relates to neuronal identity and projection patterns.

We analysed two datasets. The first consists of neurons from the entorhinal cortex in a multiplexed error-robust fluorescence in situ hybridisation (MERFISH) spatial transcriptomics dataset. The second is a single-neuron reconstruction (SNR) dataset describing projection patterns of neurons in layer 5a of the entorhinal cortex.

First, a random forest classifier was trained to examine the relationship between cell-body position and gene expression–related labels. We then applied k-means clustering to define data-driven neuronal projection types based on projection features. Next, a k-nearest neighbours classifier was used to investigate whether cell-body position can predict these projection-defined types. Finally, a pretrained SNR-based classifier was applied to MERFISH spatial coordinates to infer neuronal supertypes from projection features.

Our results suggest that cell-body position contains limited but measurable information about gene expression and brain connectivity, allowing for modest predictive performance in classification tasks.

Acknowledgments

I am grateful to my supervisors, Dr Sara Wade and Dr Amy Wilson, and to Edward Agboraw for their continuous support, guidance and valuable advice throughout this project. I would also like to thank Dr Gülşen Sürmeli for providing the data and background information that formed the basis of this work. Moreover, I want to thank my friends for their emotional support.

Generative AI Acknowledgments

Generative AI tools were used responsibly throughout this dissertation as supplementary aids to support the research and writing process. All analytical decisions, interpretations, and conclusions were made by the author. Specifically:

- ChatGPT was used to clarify unfamiliar concepts and terminology related to gene expression and neuronal projection patterns.
- ChatGPT was used to assist with debugging code and resolving technical issues encountered during data analysis and model implementation.
- ChatGPT was used to review grammar and spelling in the final report to improve clarity and readability.

University of Edinburgh – Own Work Declaration

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1 Introduction

The mammalian brain is a highly complex organ that supports a wide range of functions, including sleep, cognition, sensation, behaviour, emotion, learning, and memory[1]. Among the many brain regions involved in these functions, the entorhinal cortex is of particular interest because of its central role in memory processing. The entorhinal cortex serves as a major interface between the neocortex and the hippocampus, integrating multisensory information and supporting functions such as memory formation, spatial navigation, and temporal processing[2] Owing to its central position within memory-related neural circuits, the entorhinal cortex provides an important model system for investigating how neuronal properties relate to brain connectivity.

Previous studies have demonstrated that neuronal gene-expression patterns are associated with long-range projection targets, suggesting a close relationship between gene-expression profiles and neuronal connectivity[3]. Understanding this relationship can improve our knowledge of the mechanisms underlying brain organisation and provide insights into how neuronal genetic identity shapes connectivity patterns and memory-related neural circuits.

To investigate this relationship, we utilised two complementary multimodal datasets: a multiplexed error-robust fluorescence in situ hybridisation (MERFISH) spatial transcriptomics dataset and a single-neuron reconstruction (SNR) dataset. By integrating transcriptomic and projectomic information through cell-body coordinates, we aimed to investigate the relationship between neuronal genetic identity and brain connectivity using machine learning methods. In addition, we then focused on four specific supertypes, which are believed to best describe the layer 5a in the entorhinal cortex.

2 Data

2.1 MERFISH Spatial Transcriptomics Dataset

The transcriptomic dataset used in this study consists of 48,615 neurons from the entorhinal cortex, extracted from the MERFISH dataset. MERFISH is a spatial transcriptomics technology that profiles approximately 4.3 million cells using multiplexed error-robust fluorescence in situ hybridisation, enabling direct imaging and decoding of RNA transcripts within intact cells[1].

Conventional single-cell transcriptomic approaches typically require tissue dissociation, which results in the loss of spatial information about where each cell was originally located. In contrast, spatial transcriptomics methods preserve the anatomical context of cells while simultaneously measuring gene expression, allowing molecular profiles to be directly linked to spatial organisation within tissue[4]. This makes such approaches particularly suitable for investigating how neuronal identity is structured across anatomical space.

Each neuronal cell in this dataset is associated with gene expression profiles, spatial coordinates, and annotated cell identities. Neurons are assigned to a supertype, a transcriptomically defined cell class that groups cells with similar gene expression profiles. In this study, each neuron is represented by its 3D spatial coordinates (x, y, z), which serve as the input features, while the neuronal supertype is used as the target label. The primary objective is to investigate the relationship between spatial organisation and transcriptomic identity. Moreover, no missing values were observed in this dataset.

As shown in Figure 1, neuronal supertypes are ranked in descending order according to their sample size, with rank 1 corresponding to the most abundant supertype. The y-axis represents the number of cells in each supertype. A logarithmic scale is applied to better visualise differences across highly imbalanced classes. The results indicate a substantial class imbalance across the 167 identified supertypes.

Figure 2 reveals clear spatial structures and non-random distributions of neurons across brain regions. This suggests that spatial coordinates may contain predictive information for supertype classification.

To further explore the spatial organisation of transcriptomic identities, the spatial distributions of the ten most abundant neuronal supertypes are presented in Appendix A.1. Distinct spatial patterns can be observed across supertypes, suggesting a relationship between anatomical location and transcriptomic identity.

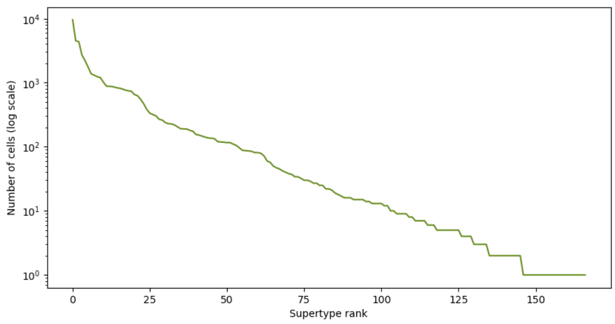


Figure 1: Rank frequency Distribution of Supertype

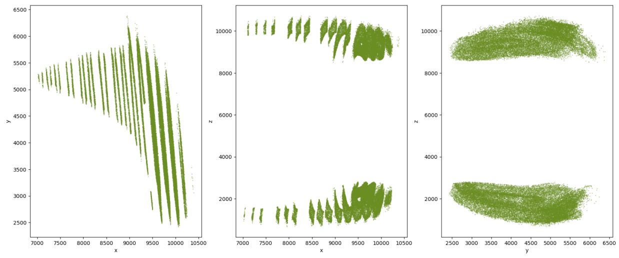


Figure 2: Cell Distribution Across K-means Clusters

2.1.1 MERFISH Preprocessing

To ensure that every supertype was represented in both the train and test sets, 21 supertypes containing only a single cell were excluded from the analysis. Excluding singleton classes reduces the risk of unstable performance estimates and ensures that each class is represented during both model training and evaluation. After this filtering step, 48,594 neurons remained for downstream analysis, with spatial coordinates (x, y, z) from MERFISH used as input features.

The dataset was then split into training and test sets prior to model fitting to avoid data leakage. The final dataset was split 80/20 into 38,875 training samples and 9,719 test samples.

2.2 Single-Neuron Reconstruction (SNR) Dataset

The Single-Neuron Reconstruction (SNR) dataset describes the projection patterns of neurons located in layer 5a of the entorhinal cortex. The dataset contains tracing information for 237 neurons obtained from 11 mouse brains. For each neuron, projection information is recorded across 29 distinct target regions. Two quantitative measures are provided for each projection area: the number of axon terminals (endpoints) observed within the target region and the total axonal length within that region. In addition, the 3D spatial coordinates (x, y, z) specify the location of each neuron's cell body within the Allen Common Coordinate Framework.

Based on their projection targets, neurons are categorised into three projection groups: neurons projecting predominantly to the orbitofrontal cortex (ORB), neurons projecting predominantly to the retrosplenial cortex (RSP), and neurons projecting to both regions.

The variables of interest in this study are the spatial coordinates of neuronal cell bodies and their corresponding projection identities. The primary objective is to investigate the relationship between cell-body position and neuronal projection patterns.

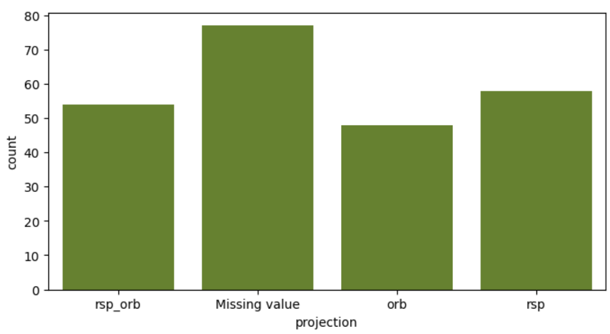


Figure 3: Distribution of neurons across projection groups

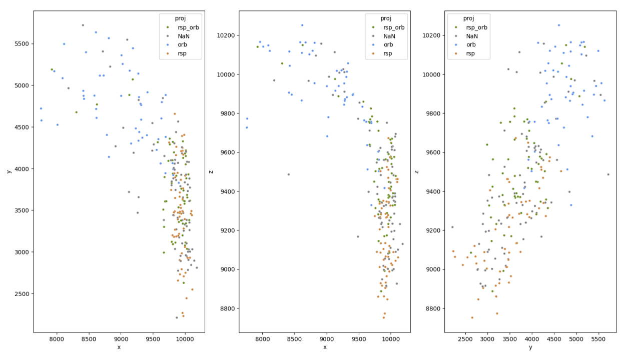


Figure 4: Spatial coordinates distribution

As shown in Figure 3, 77 neurons have missing projection annotations. These missing values reflect cases in which neurons could not be confidently assigned to a projection group, leading to incomplete annotation coverage. Given the substantial proportion of missing annotations, data-driven projection clusters were subsequently defined using unsupervised clustering on projection features. This approach enabled the identification of projection identities independently of the original annotations.

Figure 4 illustrates the spatial distributions of the three projection groups. Although substantial overlap exists among the groups, a broad spatial organisation can still be observed, with ORB-projecting neurons tending to occupy more anterior regions and RSP-projecting neurons tending to be located more posteriorly.

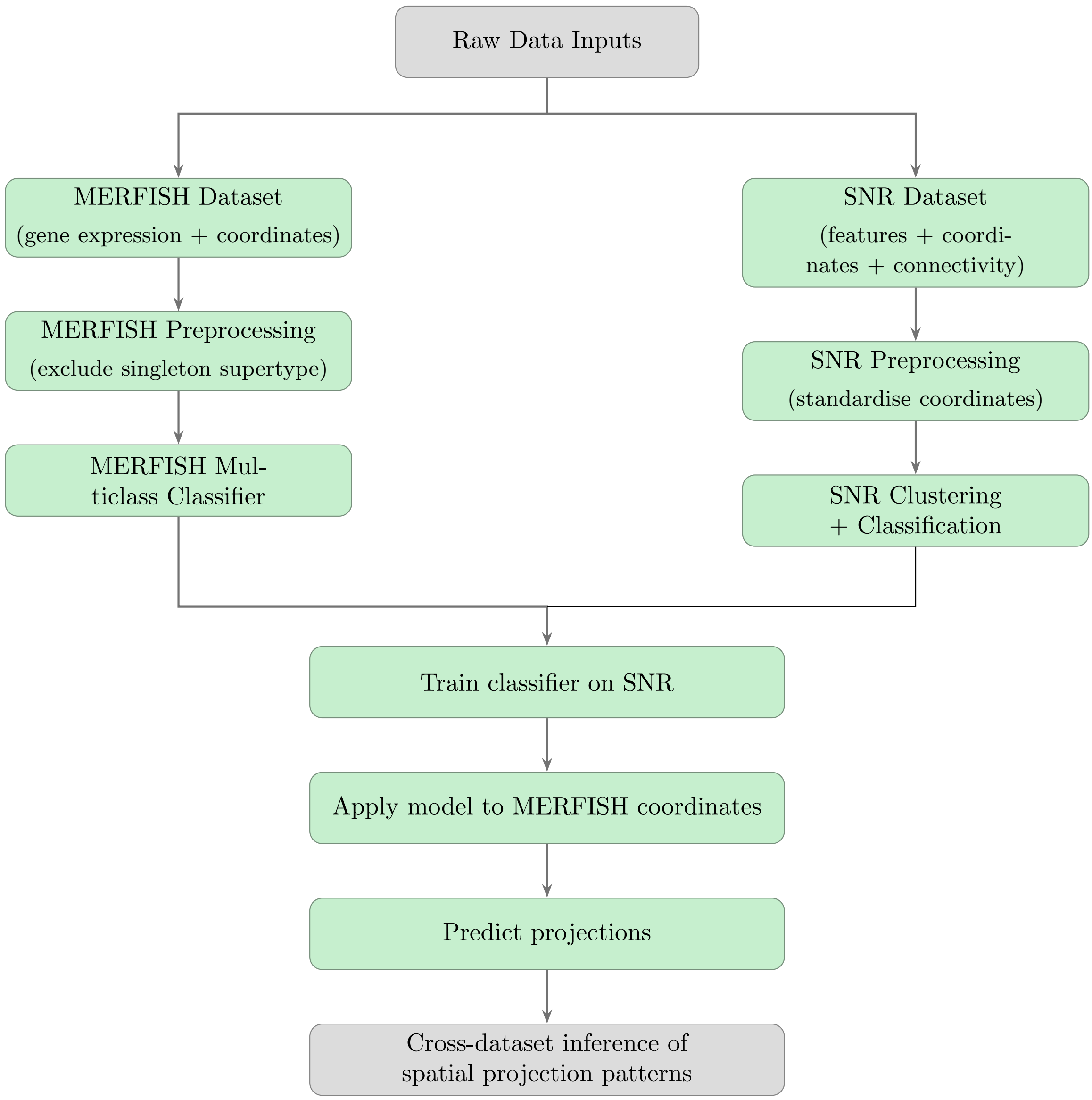
2.2.1 SNR Preprocessing

The dataset was first split into training and test sets using an 80/20 ratio, resulting in 189 training observations and 48 test observations.

The three-dimensional spatial coordinates were then standardised by centering and scaling to unit variance, with parameters estimated from the training data only and applied to both training and test sets. Standardisation was applied to ensure that spatial coordinates from different datasets were on a comparable scale, facilitating downstream cross-dataset analysis.

3 Method

3.1 Overview of the Analytical Framework



3.2 Projection cluster identification (SNR only)

Fine-grained projection classes were difficult to predict. Therefore, a clustering method, K-means, was applied to identify broader projection groups based on axonal endpoint density and total axonal length. Since K-means clustering was used as an exploratory tool to identify broad projection groups rather than to evaluate predictive performance, clustering was performed on the full dataset prior to the train–test split.

K-means clustering partitions data into K clusters by minimising the within-cluster sum of squares:

$$\arg\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2,$$

where μ_k denotes the centroid of cluster C_k. A key limitation of K-means is that the number of clusters K must be specified a priori. In this study, the optimal value of K was selected using silhouette analysis.

For clustering, two projection features were used: (i) the number of axon terminals (endpoints) within each target region and (ii) the total axonal length within the same region, measured across 29 anatomical areas. All features were standardised to zero mean and unit variance prior to clustering. K-means was then applied to the standardised feature space, and the optimal number of clusters was determined by maximising the mean silhouette coefficient.

3.3 Input and Response Definition for MERFISH and SNR datasets

For the MERFISH dataset, spatial coordinates were used as input variables for classification models. The outputs corresponded to predicted cell-type or projection-related labels depending on the task.

In the SNR dataset, projection labels were used as target variables to train a separate classifier. This classifier was subsequently applied to MERFISH data to infer projection identities.

3.4 Classification models

3.4.1 Candidate classifiers

Several classification algorithms, including a dummy classifier, a k-nearest neighbours (KNN) classifier, and a random forest classifier, were evaluated.

Dummy Classifier serves as a baseline model that ignores input features and always predicts the majority class observed in the training set. It provides a reference point to ensure that the performance of more complex models is meaningful.

k-Nearest Neighbours (KNN) Classifier assigns a label based on majority voting among the nearest training samples [5]:

$$\hat{y}(x) = \text{mode}\{y_i \mid i \in \mathcal{N}_k(x)\}$$

where $\mathcal{N}_k(x)$ denotes the set of k nearest neighbours under Euclidean distance. In this study, we set $k = 5$, which provides a balance between capturing local structure and reducing sensitivity to noise.

Random Forest Classifier were considered particularly suitable because they can model complex non-linear relationships between neuronal position and transcriptomic identity while being robust to noise. It aggregates predictions from multiple decision trees trained on bootstrap samples [6]:

$$\hat{y}(x) = \text{mode}\{T_b(x)\}_{b=1}^{B}$$

where $T_b(x)$ denotes the prediction of the b-th decision tree and B is the total number of trees in the ensemble. In our experiments, we use an ensemble of 500 trees. Increasing the number of trees generally improves stability and predictive performance by reducing variance. To address class imbalance, class weights were set to be balanced during training.

3.4.2 Model evaluation

To evaluate the performance of the classifiers, we compare a simple baseline model, a distance-based method, and an ensemble method. This setup allows us to assess whether more complex models provide improvements over simpler approaches in the classification of spatial features.

Given the highly imbalanced distribution of target variables, overall accuracy alone was not considered sufficient for model evaluation, as it can be dominated by majority classes[10]. Therefore, Balanced Accuracy and Macro F1-score were adopted as the primary evaluation metrics. Balanced Accuracy was used because it gives equal weight to each class by averaging recall across classes, making it more appropriate than overall accuracy for highly imbalanced classification problems[7]. Macro F1-score was adopted because it assigns equal importance to all classes regardless of class size, preventing performance estimates from being dominated by majority classes[9].

In addition, Top-5 Accuracy was reported in MERFISH dataset because the task involved a large number of neuronal supertypes, and biologically similar cell types may occupy nearby spatial locations[11]. Top-5 Accuracy therefore provides an additional measure of whether the true supertype is among the model's most likely predictions.

For the SNR dataset, recall was additionally reported because the binary classification task exhibited substantial class imbalance. Recall measures the proportion of minority-class samples that are correctly identified and is particularly informative when the ability to detect underrepresented classes is of primary interest.

3.5 Cross-dataset application

The best-performing classifier trained on the SNR dataset was subsequently applied to the MERFISH coordinates to investigate whether transcriptomically defined neurons exhibit projection-specific spatial signatures. Prior to prediction, the MERFISH spatial coordinates were independently standardised by removing the mean and scaling to unit variance. Predicted projection cluster labels were then summarised across neuronal supertypes to explore the relationship between transcriptomic identity and projection-associated spatial organisation. Also, to quantify the association between transcriptomic identity and projection pattern, Cramér's V was calculated. Cramér's V values always fall between 0 and 1. A value of 0 means there is no relationship between the variables, while a value of 1 indicates a perfect and complete association.

4 Result

4.1 MERFISH Classification Results

Table 1: Classification performance for predicting neuronal supertypes using MERFISH spatial transcriptomic data

Model	Accuracy	Balanced Accuracy	Macro F1	Top-5 Accuracy
Dummy Classification	0.197	0.007	0.002	0.198
K-nearest neighbours	0.566	0.114	0.107	0.732
Random forest	0.561	0.114	0.112	0.798

As shown in Table 1, compared with the dummy baseline, both k-nearest neighbour classification and random forest classification substantially improved classification performance, indicating that neuronal spatial coordinates contain information related to transcriptomic identity. The relatively high top-5 accuracy suggests that although precise prediction of individual supertypes is difficult, spatial position is nevertheless informative for identifying broader transcriptomic neighbourhoods. This finding is consistent with the notion that related neuronal classes often occupy overlapping anatomical domains. Among the evaluated models, random forest achieved the highest macro F1 score (0.112) and top-5 accuracy (0.798). However, the relatively low balanced accuracy and macro F1 scores suggest that spatial position alone provides only limited predictive power for distinguishing transcriptomic cell types.

To investigate whether classification performance was associated with spatial organization, six representative transcriptomic supertypes were selected based on their classification performance. A composite score was calculated for each supertype as

$$\text{Score} = \text{F1-score} \times \log(1 + \text{support}),$$

where support denotes the number of cells belonging to each supertype. This weighting was introduced to avoid selecting supertypes with high F1-scores but very small sample sizes, which may yield unreliable performance estimates. Consequently, the composite score favored supertypes that exhibited both strong classification performance and sufficient representation in the dataset.

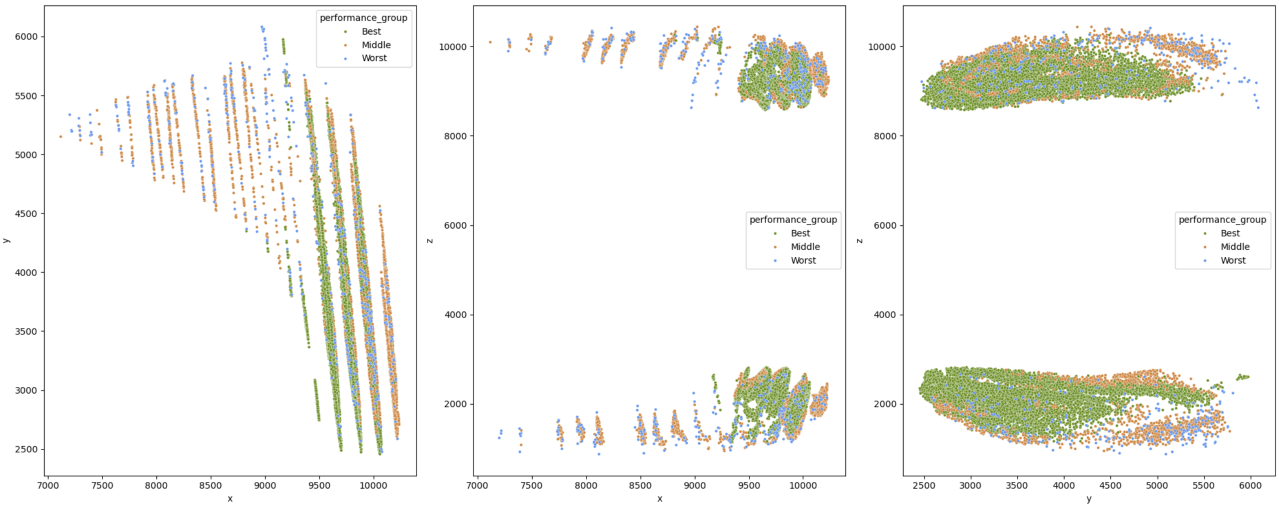


Figure 5: Spatial distribution of representative transcriptomic supertypes selected from best-, middle-, and worst-performing groups.

As shown in Figure 5, transcriptomic supertypes exhibiting high classification performance tended to form spatially localized clusters, whereas poorly classified supertypes were more broadly distributed across the coordinate space. Supertypes with intermediate classification performance displayed mixed spatial patterns, showing partial regional enrichment together with substantial overlap. These findings suggest that transcriptomic identities with stronger spatial organization are more readily predicted from neuronal cell-body position.

These findings suggest that neuronal cell-body position is associated with gene expression patterns, although spatial location alone is insufficient to accurately predict transcriptomic identity.

4.2 SNR Clustering Results

4.2.1 Selection of the Number of Clusters

As shown in Figure 6, the highest mean silhouette score was obtained at $K = 2$ (0.32), indicating that a two-cluster solution provided the best overall separation among the tested values. Alternative values of K ($K = 3$–10) were also evaluated, but all resulted in lower mean silhouette scores.

To further assess cluster quality, the distribution of sample-wise silhouette scores was examined (Figure 7). Although some overlap between clusters was observed and several samples exhibited low silhouette values, the majority of samples showed positive silhouette scores, suggesting a moderate degree of cluster separation. Overall, these results support the use of $K = 2$ for subsequent analyses.

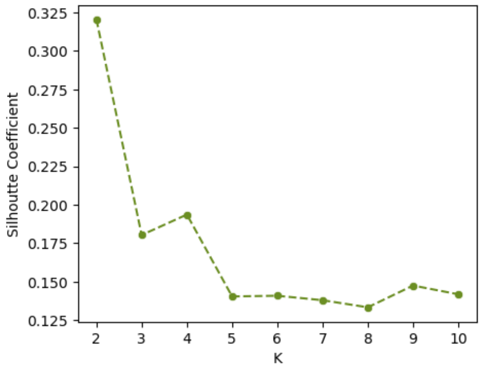


Figure 6: Mean silhouette scores for different numbers of clusters

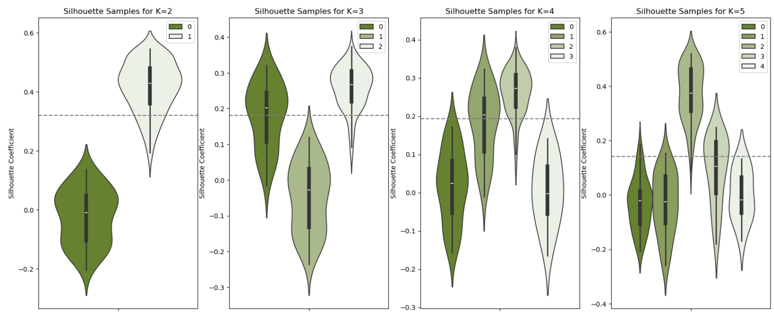


Figure 7: Distribution of sample-wise silhouette scores

4.2.2 Characteristics of Projection Clusters

Based on the optimal number of clusters ($K = 2$), we first examined the distribution of samples across clusters. As shown in Figure 8, the clustering solution exhibited a notable imbalance in cluster size, with one cluster containing substantially more neurons than the other.

We next investigated whether the two clusters differed in their anatomical projection profiles (Table 2). The two projection clusters exhibited distinct distributions along the spatial axis. Cluster 1 was concentrated at lower coordinate values, whereas Cluster 2 was predominantly located at higher values, suggesting that neuronal position is associated with projection identity.

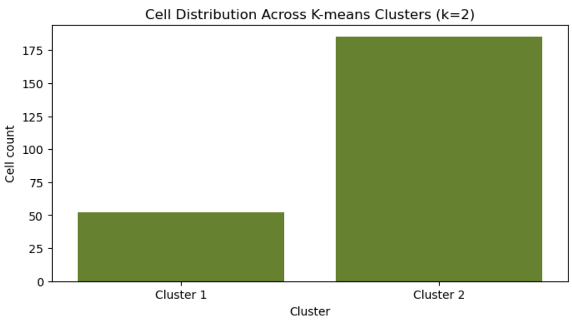


Figure 8: Distribution of projection clusters in anatomical space.

Table 2: Distribution of anatomical projection targets across clusters.

Projection Cluster	ORB	RSP	ORB_RSP	Missing
Cluster 1	32	0	16	4
Cluster 2	16	58	38	73

To further characterise these differences, we analysed the distribution of the top 10 projection areas within each cluster (Figure 9). This analysis confirmed consistent differences in projection preferences between the two clusters, with differential enrichment patterns across anatomical regions.

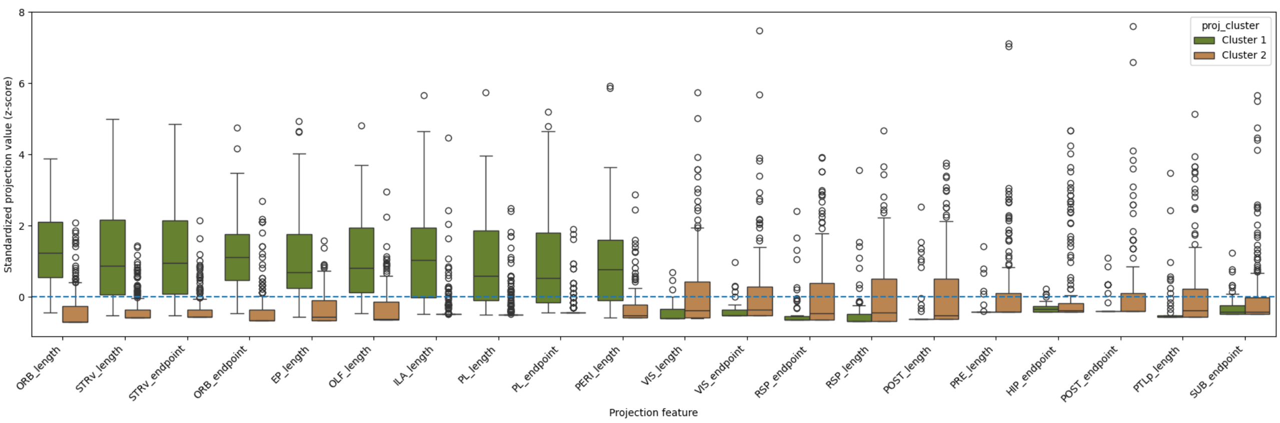


Figure 9: Distribution of top 10 area within two projection clusters

Although these patterns suggest a broad anterior–posterior organisation, we interpret Cluster 1 as anterior-biased and Cluster 2 as posterior-biased based on their relative enrichment profiles. However, substantial overlap in projection targets remained, indicating that the separation is not strict but reflects a partially mixed organizational structure.

4.3 Classification of Projection Clusters

Table 3: Classification performance for predicting projection clusters from cell-body spatial coordinates in the SNR dataset

Model	Accuracy	Balanced Accuracy	Macro F1	Recall
Dummy Classification	0.771	0.500	0.435	[0.000, 1.000]
K-nearest neighbours	0.854	0.810	0.800	[0.727, 0.892]
Random forest	0.833	0.796	0.778	[0.727, 0.865]

Compared with the dummy baseline in Table 3, both k-nearest neighbours and random forest classifiers substantially improved performance, indicating that neuronal spatial coordinates contain predictive information about projection pattern. The relatively high accuracy of the dummy classifier reflects class imbalance in the dataset, as supported by its lower balanced accuracy. Among the evaluated models, k-nearest neighbours achieved the highest balanced accuracy (0.810) and macro F1 score (0.800), slightly outperforming the random forest classifier. k-nearest neighbours was expected to perform well because neurons with similar projection identities may occupy neighbouring regions in anatomical space.

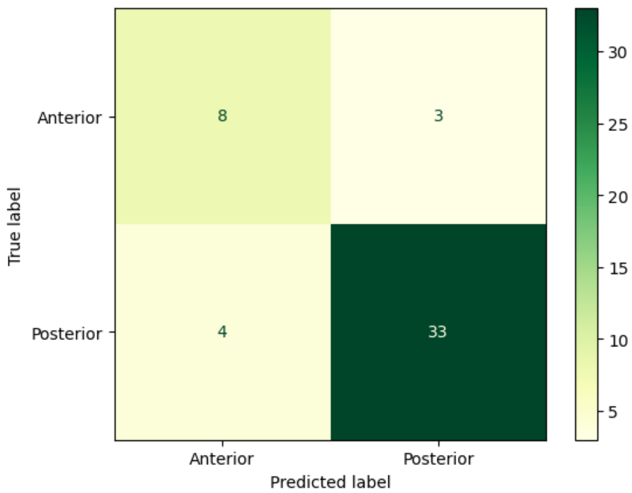


Figure 10: Confusion matrix for projection cluster classification.

As shown in Figure 10, the confusion matrix revealed asymmetric classification performance between the two projection clusters. Posterior was classified with higher accuracy (33/37 correct predictions), whereas anterior showed lower separability (8/11 correct predictions). Misclassifications were observed in both directions, indicating incomplete separability between the two clusters in the spatial feature space.

Overall, these results indicate that projection cluster identity can be partially inferred from spatial coordinates, although the separation between clusters is not complete.

4.4 Application of the SNR Classifier to MERFISH Data

By applying the SNR classifier to standardised MERFISH coordinates, each cell was assigned a predicted projection pattern. We next explored the relationship between transcriptomic identity (supertype) and brain connectivity (projection pattern).

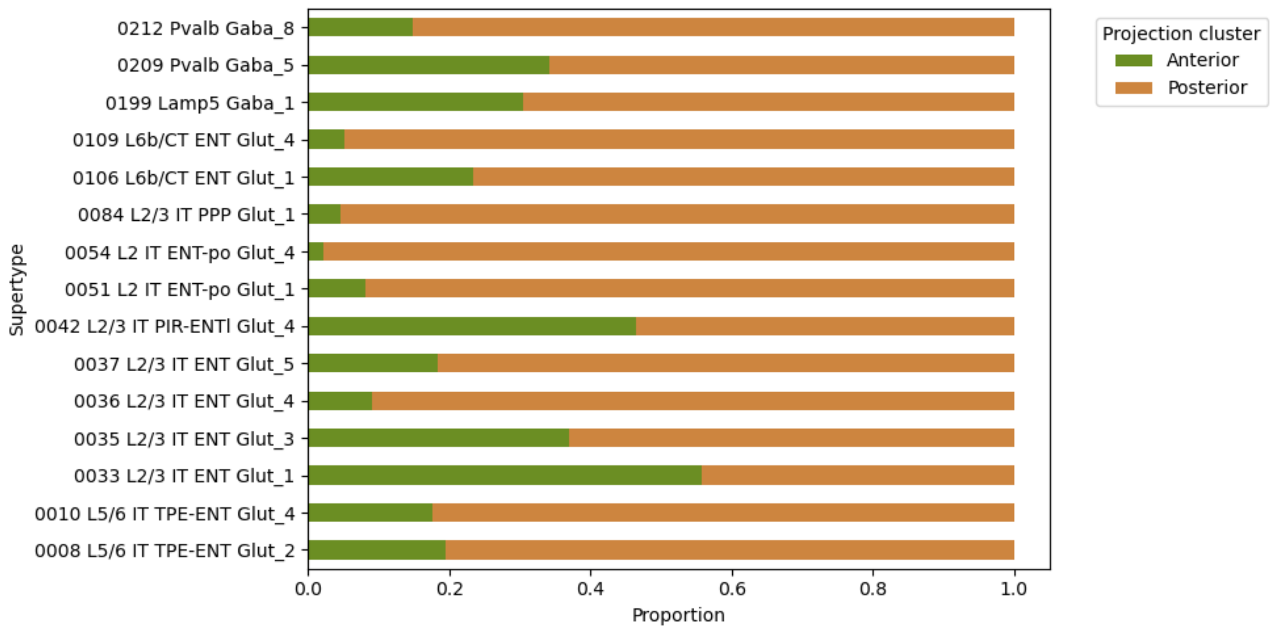


Figure 11: Projection cluster distribution across the top 15 supertypes

As shown in Figure 11, most of the top 15 supertypes were predominantly associated with posterior projection patterns. Cramér's V ($V = 0.442$) indicated a moderate-to-strong association, suggesting that genetic identity constrains connectivity organisation while still allowing substantial variability within transcriptomic classes.

These findings suggest that transcriptomic identity partially determines neuronal connectivity patterns, although considerable heterogeneity remains within individual supertypes.

4.5 Focused Analysis on Specific Supertypes

While the population-level analysis revealed an overall association between transcriptomic identity and projection patterns, a more detailed investigation was subsequently performed on four specific supertypes of particular interest. This focused analysis aimed to determine whether these selected supertypes exhibit distinct spatial distributions and connectivity characteristics. These four supertypes

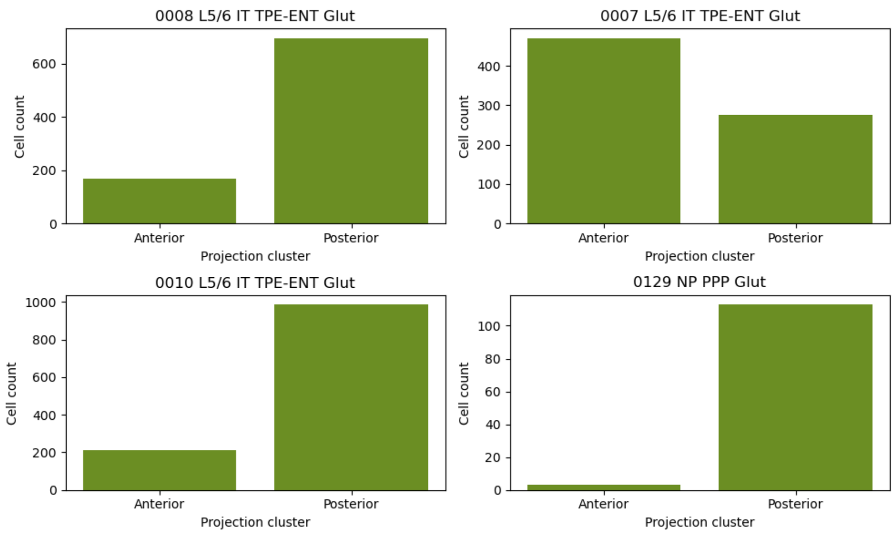


Figure 12: Projection cluster distributions of the four selected supertypes

Figure 12 shows that, although all four supertypes are believed to best describe the layer 5a in entorhinal cortex, they display distinct projection patterns. This observation suggests that neurons with similar gene expression profiles may nevertheless adopt different connectivity organisations, highlighting substantial functional heterogeneity within layer 5 neurons populations.

5 Discussion and Conclusion

The moderate association observed in this study suggests that transcriptomic identity constrains, but does not completely determine, neuronal connectivity. Additional factors, including developmental processes, neuronal activity, and local microenvironment, are likely to contribute to connectivity organisation. We also observed a moderate predictive relationship between spatial information and projection-defined clusters, indicating that neuronal position contributes to the organisation of brain connectivity.

In addition, the SNR classifier applied to MERFISH data revealed a moderate association between gene expression and connectivity. However, there was substantial heterogeneity within transcriptomic classes, with neurons of similar gene expression profiles exhibiting diverse projection patterns. This was further supported by the analysis of four specific layer 5a supertypes, which showed distinct connectivity patterns despite similar transcriptomic identities.

These findings are broadly consistent with previous studies reporting associations between transcriptomic identity and projection targets. However, the moderate rather than strong association observed here suggests that transcriptomic identity alone is insufficient to fully explain projection diversity. Overall, these results suggest that brain connectivity is shaped by both neuronal spatial position and transcriptomic identity, but neither factor alone is sufficient to fully explain connectivity organisation.

Several limitations should be acknowledged. First, connectivity labels were inferred indirectly using an SNR-based classifier rather than being directly measured for MERFISH neurons. This introduces potential uncertainty in the predicted labels and may affect downstream analyses. Second, the analysis was restricted to spatial coordinates and transcriptomic features, while other potentially informative data were not included.

From a modelling perspective, future work could incorporate additional data sources and multimodal representations to improve prediction performance and robustness. In particular, integrating spatial, transcriptomic, and connectivity-related information within a unified framework may help better capture the structure of the data and reduce classification uncertainty.

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A Appendices

A.1 Spatial Distribution of the Top 10 Most Frequent Supertypes

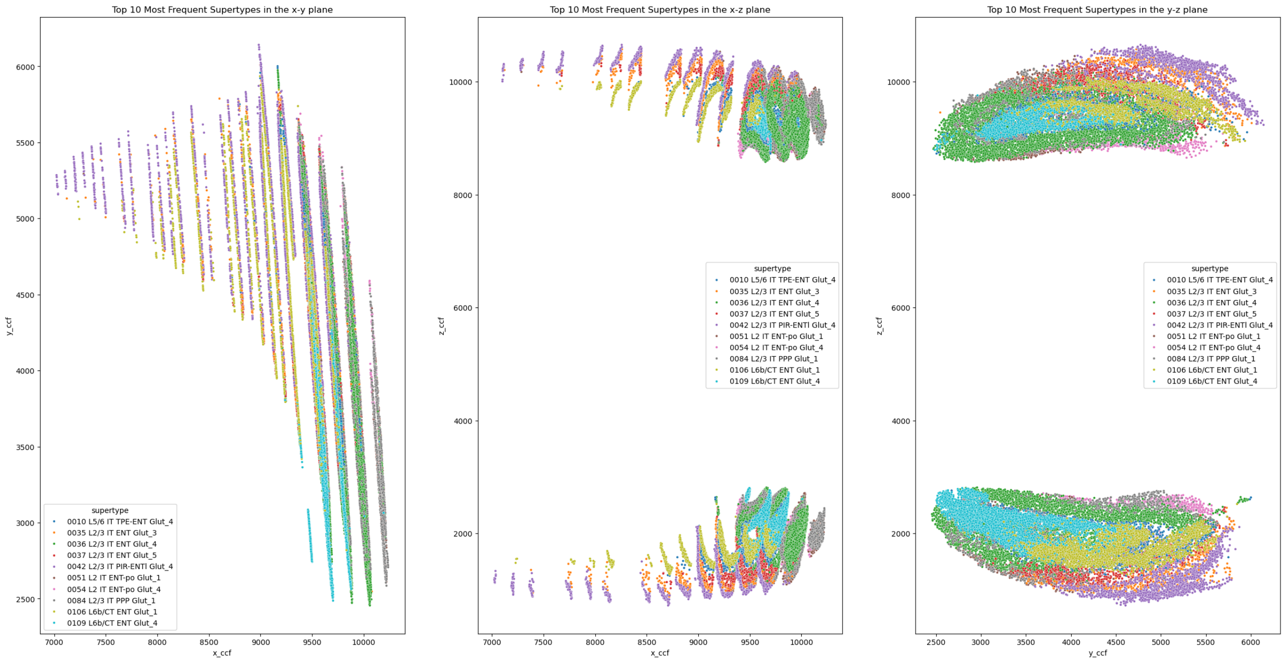

Figure 13: Spatial distributions of the ten most abundant neuronal supertypes in the x–y, x–z and y–z planes. Distinct spatial patterns can be observed across transcriptomic classes.

B Code Availability

The code for all analyses conducted in this dissertation can be found at `https://github.com/YuHsuan07/Linking-Neuronal-Genetic-Identity-to-Connectivity-Patterns-in-the` `git`. The datasets used in this study are not publicly available.